TEACHER GUIDE DRAFT

IB Diploma Programme Business management HL

Break-even analysis

Unit 5: Operations management
5.5 Break-even analysis



A revision lesson. Pupils run a car dealership in firms of four and revise break-even analysis on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Cars to break even

Round 1

They take up to 5 cars.

Cars to break even? Round up.

Livery, one-off	**£2,400**
Staff training, one-off	**£2,150**
Customer pays per car	**£18,760**
You pay per car	**£16,800**

ANSWER

£3

(£2,400 + £2,150) ÷ £1,960 = 2.32 → 3 cars

Why: Part of a car does not sell: 3 whole cars cover the costs.

2 Break-even limits

Round 1

You plan a fleet deal with a break-even chart.

Which two are limits of break-even analysis?

A It needs the firm's cash balance

B It assumes every car is sold

C It leaves out the fixed costs

D It assumes one fixed price

E It cannot show the profit

ANSWER

✓ It assumes every car is sold

✓ It assumes one fixed price

Why: It assumes all output sells at one price; it does show fixed costs and profit.

3 Safety margin as %

Round 2

Three orders, each with its break-even quantity.

Order by margin of safety as a %, lowest first?

A Order 10 cars, break-even 7

B Order 20 cars, break-even 16

C Order 8 cars, break-even 6

ANSWER

1. **Order 20 cars, break-even 16**
2. **Order 8 cars, break-even 6**
3. **Order 10 cars, break-even 7**

Why: (20 − 16) ÷ 20 = 20%, (8 − 6) ÷ 8 = 25%, (10 − 7) ÷ 10 = 30%.

Topic tasks

4 Profit on the deal

Round 2

A deal with a set-up cost.

Profit on the deal?

Livery and set-up, one-off	**£6,550**
Customer pays per car	**£12,475**
You pay per car	**£10,900**
They order	**9 cars**

ANSWER

£7,625

9 × (£12,475 − £10,900) − £6,550 = £7,625

Why: The first cars pay the set-up; the rest is profit: £7,625.

5 Which price to set

Round 3

You expect 7 sales at either price. At £9,500 a car you break even at 6; at £9,000, at 8.

Which price should you set, and why?

A £9,000 a car

B £9,500 a car

ANSWER

✓ £9,500 a car

Why: At 7 sales £9,500 clears its break-even of 6; £9,000 falls short of 8.

6 Margin after rise

Round 3

They order 8 cars.

Margin of safety now, in cars? Round break-even up.

Livery and set-up, one-off	**£7,900**
Customer pays per car	**£16,665**
Cost per car before the rise	**£13,150**
Cost rise	**10%**

ANSWER

£4

£13,150 + 10% = £14,465; £7,900 ÷ £2,200 → 4; 8 − 4 = 4

Why: The cost rise ate into the margin of safety: 4 cars left.

Final questions

Each pupil answers 6 final questions on break-even analysis. They come from these 8 questions.

1. A component sells for £48. Variable cost is £30 a unit and fixed costs are £54,000 a month. What is monthly break-even output?
   ✔ **3,000 units**

2. A repair plan costs £36 a month, variable cost £12, fixed costs £72,000. Expected sales are 4,000 plans. What is the margin of safety?
   ✔ **1,000 plans**

3. A vehicle sells for £22,500 and its variable acquisition and preparation cost is £17,100. What is unit contribution before fixed costs?
   ✔ **£5,400**

4. Monthly fixed costs rise from £90,000 to £108,000 while contribution remains £300 per vehicle. By how many vehicles does break-even output increase?
   ✔ **60 vehicles**

5. Contribution is £2,400 a vehicle at today's price; matching a rival cuts it to £1,500. Fixed costs are unchanged. Which effect follows?
   ✔ **Break-even rises: each sale covers less fixed cost**

6. A tyre service charges £120 per job, variable cost is £72 and monthly fixed costs are £38,400. What is break-even output?
   ✔ **800 jobs**

7. A service plan sells for £50 a month, has variable fulfilment cost of £18 and fixed monthly support cost of £96,000. What is break-even membership?
   ✔ **3,000 members**

8. Fixed costs are £84,000. Each vehicle sells for £16,000 and has a variable cost of £12,000. What is the break-even output?
   ✔ **21 vehicles**